Q05: Comprehension Check

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Comprehension check. A model has single-sample accuracy $p_1 = 0.5$ on a task with a *perfect* verifier. Using best-of-n, roughly how many samples n do you need so that at least one sample is correct with probability ≥ 0.95? What real-world cost grows as you do this, and name one task type where this whole strategy would *not* help.

Answer. Need $1 - (1 - 0.5)^n \geq 0.95 \Rightarrow 0.5^n \leq 0.05 \Rightarrow n \geq \log 0.05 / \log 0.5 \approx 4.32$, so $n = 5$ samples. Cost that grows: inference compute (tokens), hence latency and money—roughly linear in n. It would not help on a task with no verifier or no single "correct" answer (e.g. open-ended generation), or on a single-step lookup where p_1 is already near 1 and extra samples add nothing.